drippings, bone, subcutaneous fat, and cartilage was also weighed. Samples were homogenized and composited.

Sample analysis: Samples were chemically analyzed by qualified laboratories for nutrient content for the breast, drumstick, thigh, wing, back, and skin. Since rotisserie chicken as purchased is already cooked, samples were analyzed only as cooked, not raw. To validate all analytical procedures, quality control was monitored by use of duplicate sampling, in-house control materials, and certified reference materials.

The rotisserie chicken purchased at each individual location was used to create individual composites for analyses of proximate nutrients (moisture, total fat, ash, and protein), minerals, niacin, vitamin B6, thiamin, riboflavin, pantothenic acid, vitamin B12, cholesterol, and fatty acids. Composite samples from individual locations were grouped for analyses of selenium, folate, choline, vitamin E, amino acids, carotenoids, retinol, vitamin D, vitamin C, and vitamin K.

Whole turkey

USDA conducted a study in collaboration with Texas Tech University to determine the nutrient composition of raw and roasted whole turkey for inclusion in the USDA National Nutrient Database for Standard Reference (SR). The study was initiated due to changes occurring in the industry. For example, during sampling and market checks done by the Nutrient Data Laboratory, most whole turkeys found in retail outlets were labeled as enhanced with sodium-containing solutions, while non-enhanced whole turkeys were relatively uncommon. Therefore, due to the relatively common and recent availability of enhanced whole turkey, data for this product were needed in SR.

Sampling: Samples of whole enhanced turkeys were procured from 11 retail locations, using the nationwide sampling plan developed for the USDA's National Food and Nutrient Analysis Program (NFNAP). Due to unavailability of non-enhanced turkeys in the NFNAP retail locations, samples of non-enhanced turkeys were obtained through 4 different local retail sources. Two turkeys per location were purchased—one to be dissected and analyzed raw and the other to be dissected and analyzed after roasting. All of the turkeys were shipped to Texas Tech University for processing. Weights of meat, skin, and other components were obtained in order to determine cooking yields. Samples of meat, skin and offal (gizzard, heart, and liver) were homogenized, composited, and chemically analyzed for nutrient content.

Cooking Procedure: Roasting -Thawed turkeys were unwrapped and the weight of the drippings, neck, organ meats, and packaging were measured. Oven was preheated to 325°F/163°C. The turkeys were placed on a wire rack in a shallow roasting pan, with 1/2 cup water added in the bottom of the pan. Turkeys were roasted until they achieved an internal temperature of 165°F/74°C, when they were removed from the oven. After 20 minutes at room temperature, the cooked weight of each whole turkey was obtained.

Sample preparation: Separation of meat, skin, separable fat, cartilage, and bone- Each turkey was cut into parts: breast, wings, drumsticks, thighs, and back including the tail. Each part was weighed and refrigerated for up to 24 hours. Dissection of each turkey part was performed by carefully "scraping" the co-mingled parts with a knife blade, so that the separable fat, bone, and cartilage were separated from the meat as closely as possible, and then weighed, to measure the amount of each component.

Component weights (i.e., weights of meat, skin, separable fat, bone, and cartilage) were reported in SR. Weights of bone and cartilage were combined and reported as "refuse". For food items listed as "meat only", the skin and separable fat associated with those cuts is included in the "refuse". For food items listed "meat and skin", the skin and separable fat are considered edible and thus contribute to the nutrient profile so are not included in the refuse.

Sample composites and nutrient analyses: For each analytical sample, the turkeys purchased at each individual location were used to create an individual composite. For the enhanced turkey data, a composite of light meat and a composite of dark meat from each location were paired (n=6), then homogenized and analyzed in both raw and cooked forms. For the non-enhanced turkey data, each location was analyzed separately (n=4) for both the raw and cooked forms. Skin samples from the entire bird were pooled and analyzed both in raw and cooked forms. At this level, these nutrients were analyzed: proximate nutrients (moisture, total fat, ash, and protein), minerals, cholesterol, fatty acids, thiamin, niacin, riboflavin, vitamin B6, and vitamin B12.

Samples from two or more locations were combined to form regional or national composites for enhanced and non-enhanced turkey for these nutrients: amino acids, choline, vitamin K, folate, and retinol. Values for vitamin D, including 25-hydroxy vitamin D, were also obtained from regional composites. Nutrient analyses were performed by TTU and a commercial laboratory, whose analytical procedures were evaluated and validated through the NFNAP process. Regional composites of enhanced turkey were also analyzed for sugar and starch.

In SR, turkey listings are described as light meat, dark meat, or by specific parts. Listings in previous versions of SR also included turkey classes such as fryer, hen, or tom in some of the descriptions. However, terms used to identify specific turkey classes are no longer included in SR, since turkeys sold in the current market are not consistently identified with this information.

As a result of this study, data have been generated to create new SR items for these enhanced whole turkey items: light meat, raw and cooked (with and without skin), dark meat, raw and cooked (with and without skin), gizzard, heart, liver, neck, back, breast, wings, drumstick and thigh. Data have also been generated to update these non-enhanced whole turkey items: light meat, raw and cooked (with and without skin), dark meat, raw and cooked (with and without skin), gizzard, heart, liver, neck, breast, wings, drumstick and thigh.

Retail turkey parts

USDA conducted a study in collaboration with Texas Tech University to determine the nutrient composition of raw and roasted retail turkey parts for inclusion in the USDA National Nutrient Database for Standard Reference. The study was initiated to obtain data for these items available to consumers as an alternative to purchasing whole turkey.

Sampling: Samples of turkey drumsticks, thighs, breast, and wings were procured from 12 retail locations, using the nationwide sampling plan developed for the USDA's National Food and Nutrient Analysis Program (NFNAP). The parts in this study were not labeled as having been enhanced with sodium-containing solutions except for the breast, for which both enhanced and non-enhanced samples were procured. The parts were shipped to Texas Tech University for processing. Weights of meat, skin, and other components were obtained in order to determine cooking yields. Samples of meat and skin, both raw and cooked, were homogenized, composited, and chemically analyzed for nutrient content.

Cooking Procedure: Roasting - Thawed turkey parts were unwrapped and weighed. The oven was preheated to 350°F/176°C. The turkey parts were placed on a wire rack in a shallow roasting pan, with no water added to the pan. Thermocouples were placed in the thickest portions of the pieces. The parts were roasted, uncovered, to an internal temperature of 165°F/74°C, when they were removed from the oven. After 30 minutes at room temperature, the cooked weights were obtained.

Sample preparation: Separation of meat, skin, separable fat, cartilage, and bone- After weighing, the parts were refrigerated for up to 24 hours. Dissection of each turkey part was performed by carefully scraping the co-mingled parts with a knife blade, so that the separable fat, bone, and cartilage were separated from the meat as closely as possible then weighed, to measure the amount of each component.

Component weights (i.e., weights of meat, skin, separable fat, bone, and cartilage) are reported in SR. Weights of bone and cartilage have been combined and reported as "refuse". For items listed as "meat only", the skin and separable fat associated with those cuts is included in the "refuse". For items listed as "meat and skin", the skin and separable fat are considered edible and thus contribute to the nutrient profile so are not included in the refuse.

Sample composites and nutrient analyses: For each analytical sample, the turkey part purchased at each individual location was used to create an individual composite. For the drumsticks and wings, composites from each location were paired (n=4 or 6) then homogenized and analyzed in both raw and cooked forms. For the thighs and breast, each composite location (n=3 or 5) was homogenized and analyzed separately in both raw and cooked forms. Skin samples from the thigh and drumsticks were pooled and analyzed both in raw and cooked forms for the 'skin from dark meat' item.

At this individual composite level, proximate nutrients (moisture, total fat, ash, and protein) and minerals were analyzed.

Samples from two or more locations were combined to form regional or national composites for these nutrients: cholesterol, fatty acids, thiamin, niacin, riboflavin, vitamin B6, vitamin B12, carotenoids, and retinol. Values for amino acids and choline were also obtained from regional composites. Nutrient analyses were performed by TTU and a commercial laboratory, whose analytical procedures were evaluated and validated through the NFNAP process.

As a result of this study, data have been generated to create new SR items for enhanced turkey breast, non-enhanced turkey drumsticks, thighs, breast, and wings (with and without skin), and skin from dark meat.

Whole turkey parts study

USDA conducted a study in collaboration with Texas Tech University to determine the nutrient composition of raw and roasted whole turkey parts enhanced and non-enhanced from whole turkeys for inclusion in the USDA National Nutrient Database for Standard Reference (SR). Samples of whole enhanced turkeys and 4 non-enhanced turkeys were procured from 11 retail locations, using the nationwide sampling plan developed for the USDA's National Food and Nutrient Analysis Program (NFNAP). Whole turkey parts; breast, back, thighs, drumsticks, wings and legs were removed from whole turkeys and weighed raw and cooked prior to homogenizing and compositing. Weights of meat, skin, and other components were obtained in order to determine cooking yields. Samples of meat, skin and offal (gizzard, heart, and liver) were homogenized, composited, and chemically analyzed for nutrient content.

Sample preparation: Separation of meat, skin, separable fat, cartilage, and bone- Each enhanced and non-enhanced turkey was cut into parts: breast, wings, drumsticks, thighs, and back including the tail. Each part was weighed and refrigerated for up to 24 hours. Dissection of each turkey part was performed by carefully "scraping" the co-mingled parts with a knife blade, so that the separable fat, bone, and cartilage were separated from the meat as closely as possible, then weighed to measure the amount of each component.

Component weights (i.e., weights of meat, skin, separable fat, bone, and cartilage) were reported in SR. Weights of bone and cartilage were combined and reported as "refuse". For food items listed as "meat only", the skin and separable fat associated with those cuts is included in the "refuse". For food items listed "meat and skin", the skin and separable fat are considered edible and thus contribute to the nutrient profile so are not included in the refuse.

Cooking Procedure: Roasting - Thawed turkey parts were unwrapped and weighed. The oven was preheated to 350°F/176°C. The turkey parts were placed on a wire rack in a shallow roasting pan, with no water added to the pan. Thermocouples were placed

in the thickest portions of the pieces. The parts were roasted, uncovered, to an internal temperature of 165°F/74°C, when they were removed from the oven. After 30 minutes at room temperature, the cooked weights were obtained.

Sample composites and nutrient analyses: For each analytical sample, the turkey part purchased at each individual location was used to create an individual composite. For the drumsticks and wings, composites from each location were paired (n=4 or 6) then homogenized and analyzed in both raw and cooked forms. For the thighs and breast, each composite location (n=3 or 5) was homogenized and analyzed separately in both raw and cooked forms. Skin samples from the thigh and drumsticks were pooled and analyzed both in raw and cooked forms for the 'skin from dark meat' item. At this individual composite level, proximate nutrients (moisture, total fat, ash, and protein) and minerals were analyzed.

Samples from two or more locations were combined to form regional or national composites for these nutrients: cholesterol, fatty acids, thiamin, niacin, riboflavin, vitamin B6, vitamin B12, carotenoids, and retinol. Values for amino acids and choline were also obtained from regional composites. Nutrient analyses were performed by TTU and a commercial laboratory, whose analytical procedures were evaluated and validated through the NFNAP process.

As a result of this study, data have been generated to create new SR items for enhanced and non-enhanced turkey breast, drumsticks, thighs, back, wings and legs.

Enhanced and non-enhanced dark meat chicken

USDA conducted a study in collaboration with Texas Tech University to determine the nutrient composition of raw and cooked chicken drumsticks and thighs sold as retail parts, for inclusion in SR. Samples of non-enhanced dark meat chicken (n=7) and enhanced (n=7) were procured from 12 retail locations, using the nationwide sampling plan developed for the USDA's National Food and Nutrient Analysis Program (NFNAP). Two packages of thighs and drumsticks per location were purchased from retail stores —one to be dissected and analyzed raw and the other to be dissected and analyzed after roasting and braising. All of the chicken thighs and drumsticks were shipped to Texas Tech University for processing. Weights of meat, skin, and other components were obtained in order to determine cooking yields. Samples of meat and skin were homogenized separately, composited, and chemically analyzed for nutrient content.

Cooking Procedures:

Roasting - Chicken drumsticks and thighs were weighed. The oven was preheated to 350°F/176°C. The drumsticks and thighs were placed on a wire rack in a shallow roasting pan, with no water added to the pan. Thermocouples were placed in the thickest portions of the pieces. The drumsticks and thighs were roasted, uncovered, to an internal temperature of 165°F/74°C, when they were removed from the oven. After 30 minutes at room temperature, the cooked weights were obtained.

Braising - Oven was preheated to 325°F/ 163°C. The chicken drumsticks and thighs were weighed and placed on a roasting pan. Distilled water (100 ml) was added to the roasting pan, which was covered tightly and placed in the center of the oven. Cooking time was determined from initial trials. Initial cooking time estimates were 45 minutes for drumsticks and thighs. The internal temperature was determined with an electronic digital thermometer. Drumsticks and thighs were allowed to cool for 5 minutes and then re-weighed; weights were recorded.

Sample preparation: Separation of meat, skin, separable fat, cartilage, and bone: After weighing, the drumsticks and thighs were refrigerated for up to 24 hours. Dissection of each was performed by carefully scraping the co-mingled parts with a knife blade, so that the separable fat, bone, and cartilage were separated from the meat as closely as possible then weighed, to measure the amount of each component.

Component weights (i.e., weights of meat, skin, separable fat, bone, and cartilage) are reported in SR. Weights of bone and cartilage have been combined and reported as "refuse". For items listed as "meat only", the skin and separable fat associated with those cuts is included in the "refuse". For items listed as "meat and skin", the skin and separable fat are considered edible and thus contribute to the nutrient profile so are not included in the refuse.

Sample composites and nutrient analyses: For each analytical sample, the drumstick and thigh purchased at each individual location was used to create an individual composite. For the drumsticks and thighs, composites from each location were paired (n=4 or 6) then homogenized and analyzed in both raw and cooked forms. Skin samples from the thigh and drumsticks were pooled and analyzed both in raw and cooked forms for the "skin from dark meat" item. At this individual composite level, proximate nutrients (moisture, total fat, ash, and protein) and minerals were analyzed.

Samples from two or more locations were combined to form regional or national composites for these nutrients: cholesterol, fatty acids, thiamin, niacin, riboflavin, vitamin B6, vitamin B12, carotenoids, and retinol. Values for amino acids and choline were also obtained from regional composites. Nutrient analyses were performed by TTU and a commercial laboratory, whose analytical procedures were evaluated and validated through the NFNAP process.

As a result of this study, data have been generated to create new SR items for enhanced chicken drumsticks and thighs, non-enhanced chicken drumsticks and thighs plus skin.

Enhanced and non-enhanced light meat chicken study

USDA conducted a study in collaboration with Texas Tech University to determine the nutrient composition of raw and cooked enhanced and non-enhanced skinless, boneless chicken breast and wings with skin sold as retail parts, for inclusion in SR.

Samples of non-enhanced skinless, boneless breasts (n=12) and enhanced (n=12) plus non-enhanced chicken wings with skin (n=12) were procured from 12 retail locations, using the nationwide sampling plan developed for the USDA's National Food and Nutrient Analysis Program (NFNAP). A package of enhanced and non-enhanced breast and non-enhanced wings with skin per location were purchased from retail stores. Half of the package was dissected and analyzed raw and the other to be dissected and analyzed after roasting, braising and grilling. All of the chicken breasts and wings were shipped to Texas Tech University for processing. Weights of meat, skin, and other components were obtained in order to determine cooking yields. Samples of meat and skin were homogenized separately, composited, and chemically analyzed for nutrient content.

Cooking Procedures:

Roasting - Chicken wings with skin were weighed. The oven was preheated to 350°F/176°C. The wings were placed on a wire rack in a shallow roasting pan, with no water added to the pan. Thermocouples were placed in the thickest portions of the pieces. The wings were roasted, uncovered, to an internal temperature of 165°F/74°C, when they were removed from the oven. After 30 minutes at room temperature, the cooked weights were obtained.

Braising - Oven was preheated to 325°F/163°C. The chicken breasts were weighed and placed on a roasting pan. Distilled water (100 ml) was added to the roasting pan, which was covered tightly and placed in the center of the oven. Cooking time was determined from initial trials. Initial cooking time estimates were 45 minutes for the breasts. The internal temperature was determined with an electronic digital thermometer. Chicken breasts were allowed to cool for 5 minutes and then re-weighed; weights were recorded.

Sample composites and nutrient analyses: For each analytical sample, the chicken breast and wings purchased at each individual location was used to create an individual composite. For the breast and wings, composites from each location were paired (n=4 or 6) then homogenized and analyzed in both raw and cooked forms. At this individual composite level, proximate nutrients (moisture, total fat, ash, and protein) and minerals were analyzed.

Samples from two or more locations were combined to form regional or national composites for these nutrients: cholesterol, fatty acids, thiamin, niacin, riboflavin, vitamin B6, vitamin B12, carotenoids, and retinol. Values for amino acids and choline were also obtained from regional composites. Nutrient analyses were performed by TTU and a commercial laboratory, whose analytical procedures were evaluated and validated through the NFNAP process.

As a result of this study, data have been generated to create new SR items for enhanced and non-enhanced, boneless, skinless chicken breasts and wings with skin.

Ruffled Grouse and Canadian Goose

A study was conducted in collaboration with Cornell University to determine the nutrient composition of ruffed grouse (n=2) and Canadian goose species (n=2). Both of these species were obtained from different locations in Minnesota, New York, and Vermont. Protocols on field dressing, fabrication, and dissection were provided to Cornell University personnel, who oversaw the study. Weights for component factors such as separable lean, bone and connective tissue (refuse) were determined at Cornell University. Nutrient analysis for proximates, minerals and fatty acids on the edible meat were conducted at Texas Tech University. Nutrient values for raw meat without skin were added to SR for each of these species.

References for Notes on Foods – Poultry Products

Jones DB. Factors for converting percentages of nitrogen in foods and feeds into percentages of protein. US Department of Agriculture, Circular 83, slight revision, 1941.

Merrill AL, Watt BK. Energy value of foods: Basis and derivation, revised. US Department of Agriculture, Agriculture Handbook 74, 1973.

Perry CR, Pehrsson PR, Holden JM. A revised sampling plan for obtaining food products for nutrient analysis for the USDA national nutrient database 2003. Proceedings of the American Statistical Association, Section on Survey Research Methods [CD-ROM]. 2003. Alexandria, VA: American Statistical Association, San Francisco.

Vegetable and Vegetable Products (Food Group 11)

The sources of nutrient data for this food group include the scientific literature, analytical studies conducted by NDL and other government agencies, and the food industry. Since the inception of NFNAP (p. 52) in 1997, a few canned and frozen vegetables have also been sampled and analyzed. Data for raw and cooked vegetables, with the exception of the studies described below, for the most part, come from the scientific literature. In most cases, data for other canned and frozen vegetables were supplied by the food industry.

In 2000 and 2001 the Produce for Better Health Foundation collaborated with the Nutrient Data Laboratory (NDL) to sample and analyze 16 vegetables utilizing NFNAP protocols (p. 52). For most vegetables, samples were collected from retail outlets in 12 major metropolitan areas, once during the peak market season for the particular vegetable and once during the low market season. Four composites comprised of sample units from three locations each were prepared, and sent to the lab for analysis. A number of vegetables (tomatoes, potatoes, broccoli, and broccoli raab) were also analyzed both raw and prepared. In most cases, vegetables in the retail market are not identified by cultivar. However, for lettuce, red leaf, green leaf, iceberg, romaine, and butterhead types were sampled; for potatoes, russet, red and white types were sampled. Since 2002, additional data were obtained via NFNAP for other vegetables on FDA's list of the 20 most commonly consumed fruits, nuts and vegetables (FDA/DHHS, 2011).

In 2004 the Mushroom Council collaborated with NDL to update the data on several types of mushrooms (white, maitake, oyster, enoki, shiitake, crimini and portabella). These samples were also collected from retail outlets in 12 major metropolitan areas. They were composited into four composites of samples from three locations each, and sent to the labs for analysis. Samples of white and portabella mushrooms were also cooked before analysis. In 2009 additional analyses were conducted for vitamin D in the above listed mushrooms. Samples of morel and chanterelle mushrooms were also collected for the first time.

Data are presented for raw, canned, dehydrated, and frozen vegetables and vegetable products. When appropriate, data are presented for both the unprepared and the prepared forms of the food.

Nutrient data for different forms of a vegetable were not necessarily derived from the same sample. That is, a single sample of snap beans was not analyzed in each of the three forms: raw, canned, and frozen. The data were obtained from many sources and may represent different growing years, growing areas, cultivars, processing techniques, lengths and conditions of storage, laboratories, and possibly different methods of analysis. Therefore, in a comparison of different forms of a vegetable, nutritional differences should not be ascribed solely to the effect of processing or preparation methods.

Several factors such as natural variation, differences in postharvest handling and storage, and variations in the processing or preparation method may cause a processed or prepared form of a vegetable to have a higher nutrient content than the unprocessed or unprepared form. The differences in nutrient content between the combined solids and liquid portion and the drained solids portion of canned vegetables may not accurately reflect the amount in the drained liquid portion for the reasons stated above.

Ascorbic acid in fresh vegetables such as cabbage, lettuce, and squash is extremely unstable and is easily oxidized, particularly after chopping, cutting, or shredding. This can be similar to what occurs during the preparation of the samples for analysis. Ascorbic acid is found in foods in the reduced form, and can be converted reversibly to dehydroascorbic acid. Both the reduced and the dehydroascorbic acid forms have vitamin activity. Dehydroascorbic acid can be oxidized to diketogulonic acid, which has no vitamin activity.

Potatoes are frequently stored for many months, which may cause a decrease in ascorbic acid content. The values in the database reflect an average of several typical storage times used commercially.

The vitamin A content of vegetables such as carrots and sweet potatoes increases as they mature. Canned and frozen forms of carrots or sweet potatoes are often more mature than fresh forms, utilize different cultivars, and thus may have higher vitamin A values. Sweet potatoes have been bred in recent years to develop cultivars with deep orange—colored flesh, and have, at the same time, increased in their vitamin A content. Data collected for the database reflect the considerable variability between different samples due to natural variation and differences in methodology.

The carbohydrate content of starchy vegetables, such as sweet corn, varies with maturity, cultivar, and length of storage. As the kernels mature, moisture decreases and the carbohydrate content increases, causing a corresponding increase in calories.

A large portion of the total carbohydrates in globe or French artichokes and Jerusalem-artichokes may be inulin, a naturally occurring fructose polysaccharide, which is often measured by some analytical methods as dietary fiber. During storage, inulin is converted to available sugars (Watt and Merrill, 1963). Consequently, as the carbohydrates change with storage, the energy value also varies. The values in the tables are for the stored forms of artichokes and Jerusalem-artichokes.

Federal regulations (US Food and Drug Administration—Department of Health and Human Services, 2004) allow for the addition to canned vegetables of certain calcium salts as firming agents. The standards specify how much of these salts can be added. The addition of these salts is reflected in the values in the database.

Oxalic Acid.—Oxalic acid can combine with calcium and magnesium to form highly insoluble compounds which can make these minerals unavailable to the body. However, most foods do not contain enough oxalic acid to combine with a significant amount of